

Mammoth Cave National Park

STATEMENT OF WORK

Cyclic Repainting of Elevator Structure and Replace Deteriorating Lift Cables

PMIS# 191155

DATED: 12/3/2025



NATIONAL PARK SERVICE

SOUTHEAST REGION

PART 1 – GENERAL

ITEM 1 – DESCRIPTION OF WORK

- 1.1 The work of this contract includes the complete repainting of the 270-foot-high steel elevator support structure, and the replacement of all deteriorated elevator lift cables within the vertical elevator shaft at Mammoth Cave National Park. The work encompasses:
- Cleaning and preparing all structural steel framing, ladders, landings, and other exposed steel elements within the elevator shaft.
 - Applying rust-inhibitive primer and high-performance epoxy topcoats to all prepared surfaces.
 - Installation of new elevator lift cables conforming to NEII and ANSI A17.1 code requirements.
 - Implementation of ventilation, containment, environmental, and worker safety controls in a damp and enclosed vertical shaft environment.
- 1.2 Due to the elevator's location within a cave environment, the project is technically complex and will require stringent adherence to environmental controls, air quality, and worker safety per OSHA 1926, NFPA 101, and NPS cave protection standards.

ITEM 2 – LOCATION

- 2.1 The work is located within the vertical elevator shaft of Mammoth Cave National Park's primary visitor elevator. Access is via the surface Elevator Building and limited catwalks. The shaft descends approximately 270 feet into the cave system.

ITEM 3 – CONTRACTOR'S USE OF PREMISES

- 3.1 Contractor must confine operations to designated areas.
- 3.2 The elevator shaft will be closed to public and park staff during construction.
- 3.3 Staging, storage, and access will be coordinated with the COR.
- 3.4 Protective barriers must be installed at all access points, including stationing a guard at open elevator doors, or with the installation of locked barriers.
- 3.5 Damage to cave walls, formations, or park facilities must be prevented and, if occurs, fully remediated at Contractor's expense.

ITEM 4 – WORK SEQUENCE

- 4.1 Mobilization, access setup, containment
- 4.2 Lift cable removal and replacement (first priority)

- 4.3 Surface prep and rust treatment of structural steel
- 4.4 Application of primer and finish epoxy coatings
- 4.5 Final system testing and site restoration
- 4.6 All work must proceed in a logical, phased manner to avoid rework or unsafe conditions, with daily oversight by the Contractor's on-site superintendent.

ITEM 5 – EXISTING CONDITIONS

- 5.1 The shaft is damp, confined, and may exhibit elevated humidity and condensation.
- 5.2 Existing cables are at the end of their lifecycle and must be removed.
- 5.3 The existing structural steel may show corrosion and scaling; detailed inspection and profile prep are required.
- 5.4 Temporary drying or dehumidification may be required.
- 5.5 Any observed structural damage, including cracked welds, deformed steel, or compromised connections, must be immediately reported to the COR for assessment.
- 5.6 No structural repairs shall proceed without prior written approval. If Hot Work (e.g., welding, grinding, cutting) is required, the contractor shall implement a Hot Work Permit system, including fire watch, fire-resistant blankets, gas detection, and compliance with NFPA 51B, OSHA 1926 Subpart J, and park-specific fire protection protocols.

ITEM 6 – SUBMITTALS

- 6.1 Submit the following 15 days prior to mobilization:
 - Surface prep and painting plan (SSPC, OSHA compliant)
 - Containment, vapor barrier, and ventilation plan (LEL <10%)
 - Lift cable specifications (NEII/ANSI compliant)
 - Safety plan including fall protection, egress, and rescue
 - Work schedule with durations, milestones, dependencies
 - Equipment and paint product data sheets
 - Qualifications of painting and elevator subcontractors

ITEM 7 – TEMPORARY FACILITIES AND CONTROLS

- 7.1 Contractor to provide all sanitary, electrical, and water needs.
- 7.2 No fuel-powered equipment allowed in the shaft.

7.3 Use only low-VOC, OSHA-compliant, explosion-proof equipment.

7.4 Electrical must meet NEC, OSHA 1926.405, and be grounded <25 ohms.

7.5 Contractor to install negative-pressure ventilation with HEPA and LEL monitoring.

ITEM 8 – SAFETY REQUIREMENTS

8.1 OSHA 29 CFR 1926 compliant fall protection, scaffolding, and confined space protocols are mandatory.

8.2 All personnel must be trained in elevator shaft safety.

8.3 Daily safety toolbox talks, weekly formal safety meetings.

8.4 The shaft is a hard-hat and respirator-mandatory zone.

8.5 Emergency egress plans must be posted at all access points.

ITEM 9 – PROJECT MEETINGS

9.1 Preconstruction meeting with CO and COR mandatory.

9.2 Weekly progress meetings required; new subs must attend orientation.

9.3 Daily briefings before shift start covering hazards and work zones.

ITEM 10 – PROJECT CLOSEOUT

10.1 Final cleaning of shaft, surface restoration, and demobilization.

10.2 Submit:

- Punch list completion cert
- As-built lift cable logs
- Paint product batch reports
- Vapor test logs
- Before/after photo documentation
- Warranty documents
- Certified test results verifying lift cable tensile strength and conformance with ANSI A17.1 and NEII standards, stamped by the installing elevator contractor. Tests must include pull force, elongation, and inspection for defects or manufacturing irregularities

ITEM 11 – REFERENCES AND STANDARDS

- 11.1 Painting and Surface Prep: SSPC-SP1, SP2, SP3, SP7, SP15; ASTM D4285, D4417, D3359, D4541
- 11.2 Elevator Components: NEII-1, ANSI A17.1, ASME A17.5
- 11.3 Safety and Access: OSHA 29 CFR 1926, NFPA 101, IBC
- 11.4 Environmental Protection: EPA 832-R-92-005, NPS DO#13A
- 11.5 Fall Protection: 29 CFR 1926.500–503
- 11.6 Electrical Safety: NEC (NFPA 70), OSHA 1926 Subpart K
- 11.7 Confined Space: 29 CFR 1910.146

ITEM 12 – PRODUCTS

12.1 2.1 Painting System

- Primer: Noxyde Red, 20 mil wet application
- Finish: Noxyde Federal Gray, 20 mil wet application
- Application Method: Airless spray, brush for edges/stripes
- Film Thickness: Total 14 mils DFT minimum
- VOC Compliance: Low-VOC, <100 g/L, with MSDS submittals

12.2 Elevator Lift Cables

- Provide new traction steel elevator cables sized per existing sheaves and car weight.
- Must meet or exceed ANSI A17.1-2019 and NEII requirements.
- Must be installed and tested by a certified elevator contractor.
- Include termination hardware, thimbles, wedge sockets, and dampers.

ITEM 13 – EXECUTION

13.1 Mobilization and Access Control

- Install barriers and access control at top/bottom elevator entrances.
- Set up negative-pressure containment and ventilation per SSPC Guide 6.
- Implement vapor monitoring system to maintain LEL <10%.
- Clean all tools and equipment before cave entry (EO 13112 invasive species control).

13.2 **Removal of Elevator Cables**

- Lockout-tagout system per OSHA 1926.417.
- Disconnect, lower, and remove all existing cables under controlled hoisting.
- Inspect sheaves, drums, and guides for wear and corrosion.
- Document all removals with date-stamped photos.

13.3 **Cable Installation**

- Install new cables per manufacturer specs, using calibrated tension meters.
- Verify alignment, load balance, and damping.
- Perform operational testing and document safety certification.

13.4 **Surface Preparation**

- Clean surfaces with Cleaner/Degreaser and fresh water rinse.
- Use SSPC-SP2/SP3 for general areas; SSPC-SP7 for heavily rusted areas.
- Profile to be confirmed with replica tape (ASTM D4417, Method C).
- Document air cleanliness with ASTM D4285.

13.5 **Painting**

- Apply stripe coats to all edges, welds, and fasteners.
- Apply primer and topcoat to required DFT.
- Check wet film thickness per ASTM D4414 during application.
- Dry film measured with magnetic gauges per SSPC-PA 2.

13.6 **Environmental and Shaft Protection**

- All fumes must be vented upward through mechanical systems; natural convection allowed only when <45°F.
- Protect cave walls from overspray using impermeable tarps.
- Seal elevator bottom door with airtight vapor barrier and inspect daily.
- Use silt traps and pump filters to prevent paint or rinse water from entering cave drainage systems.

13.7 **Daily Inspections and Logs**

- Maintain log of LEL readings, ambient conditions, worker entries/exits.
- Paint batch logs, DFT readings, cable tension reports must be submitted weekly.
- The COR will conduct visual and destructive testing (Tooke gauge) as needed.

13.8 **Structural Repairs and Hot Work**

- If during cleaning or painting operations any structural deficiencies are identified, such as broken welds, fatigue cracks, or deformation, the Contractor shall:
 - Immediately notify the COR with photographic evidence and location mapping.
 - Suspend work in the affected area until directed otherwise.
 - Submit a written repair plan stamped by a licensed structural engineer, including method of repair, materials, and QA/QC checkpoints.
- Hot Work operations shall only proceed under a site-specific Hot Work Permit Program conforming to:
 - NFPA 51B: Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
 - OSHA 1926.352–354: Fire protection and welding standards
 - NPS/park-specific fire and confined space procedures.
- Fire watch personnel and extinguishers must be present for all Hot Work. No work may proceed without written approval from the COR.

13.9 **Final Cleanup and Demobilization**

- Remove all masking, barriers, debris.
- Clean shaft base of all residues.
- Submit as-built documentation, cable certifications, and final inspection checklist for COR review.

ITEM 14 – PICTURES

Overview of Elevator Building



Primary egress to Elevator Building



Back of Elevator Building



Elevator Door (Topside)



Elevator Floor Select Panel (Interior elevator)



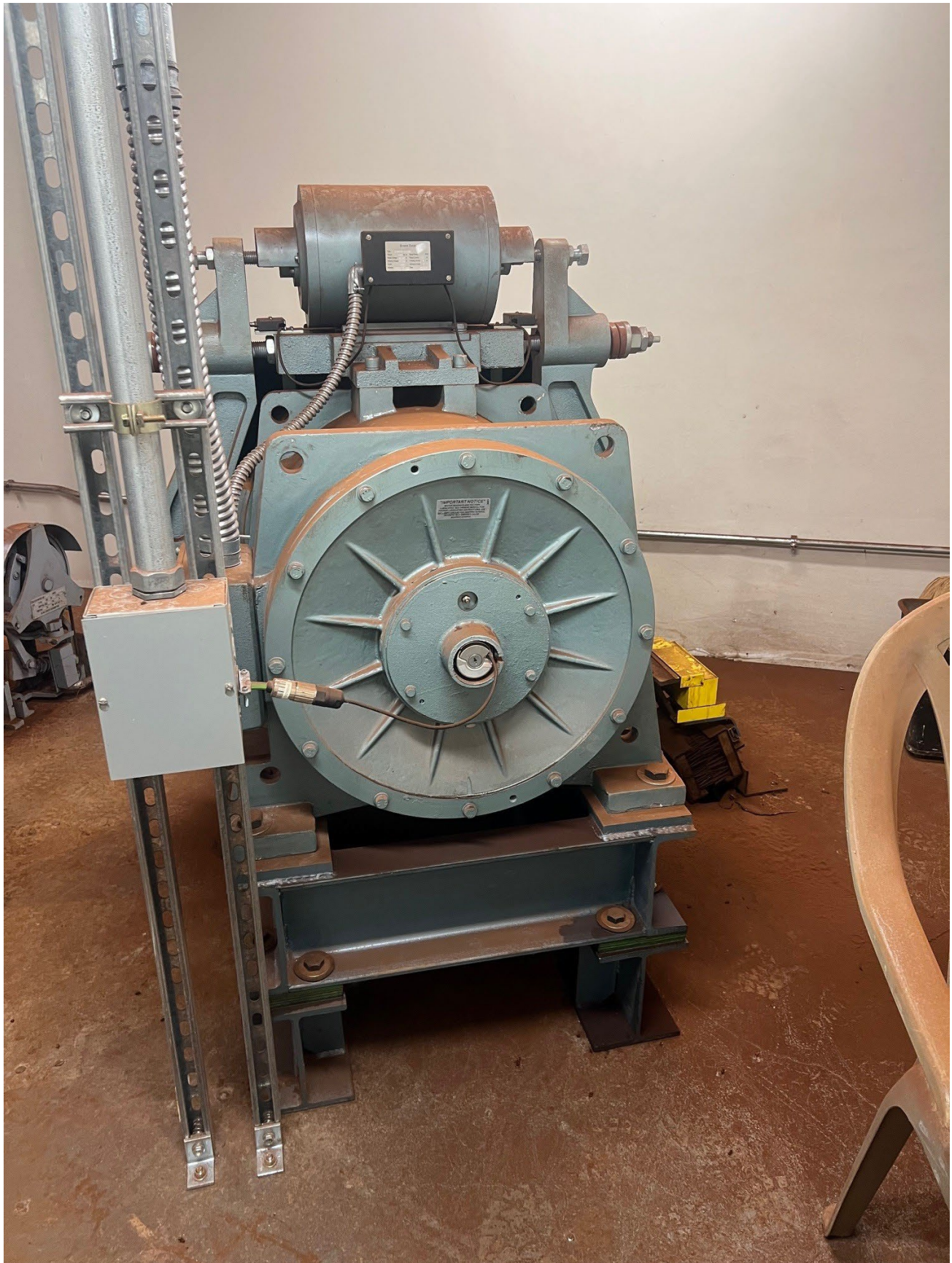
Electrical Control Panel (Access floor above elevator – Topside)



Electrical Control Cabinet (Access floor above elevator – Topside)



Elevator Drive Mechanism (Access floor above elevator – Topside - Front)



Elevator Drive Mechanism (Access floor above elevator – Topside - Side)



Elevator Drive Mechanism (Access floor above elevator – Topside - Back)

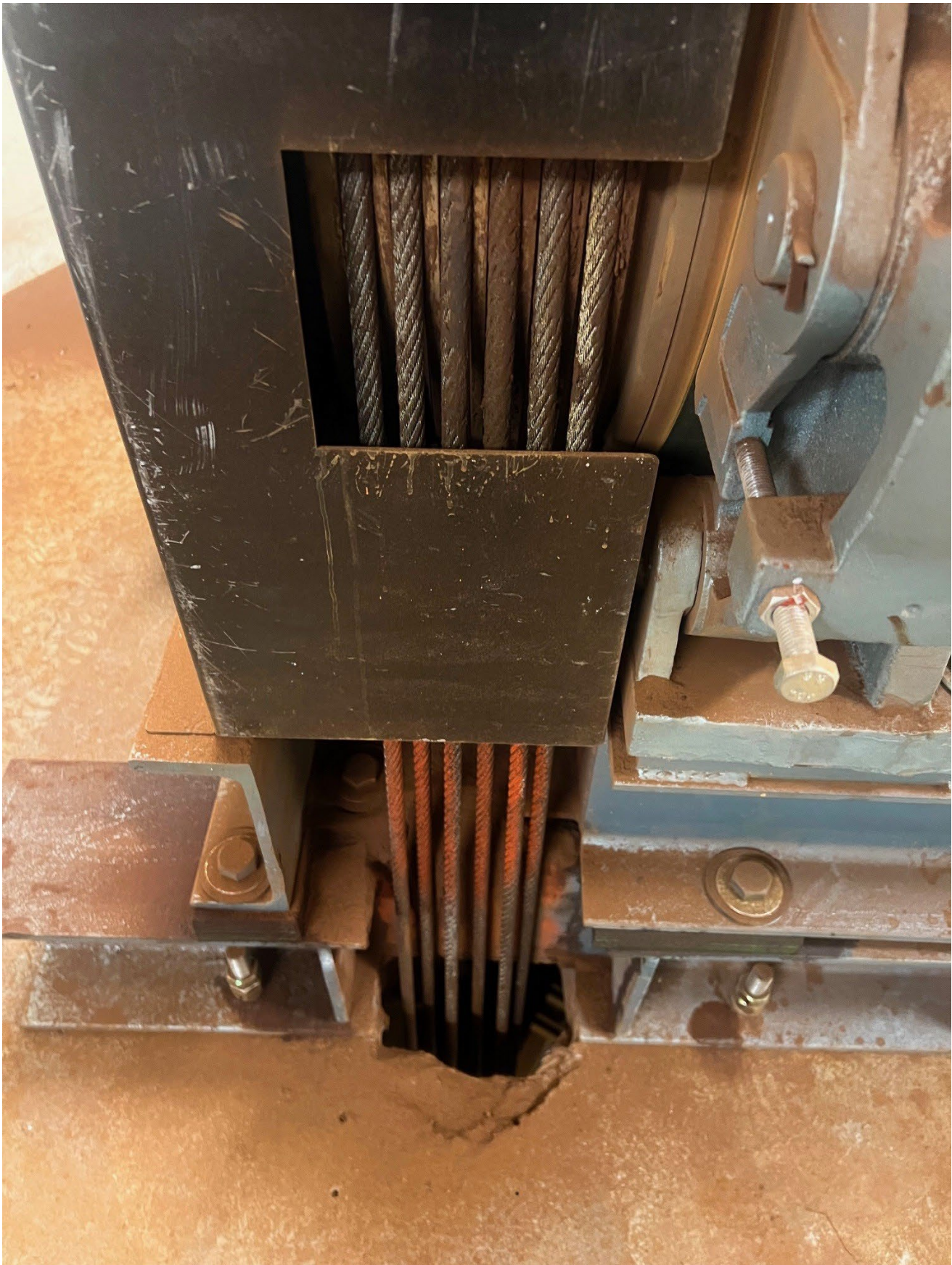


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Old Elevator Cables - Front



Old Elevator Cables - Back



Elevator Shaft and Structure (1)



Elevator Shaft and Structure (2)



Elevator Shaft and Structure (3)



Elevator Shaft and Structure (4)

